NING YU

TACTILE PERCEPTION & APPLICATION · REINFORCEMENT LEARNING · IMITATION LEARNING · VLA & WM · SYSTEM IDENTIFICATION

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EDUCATION & TRAINING

Dalian University of Technology (985) 2024.09 – 2027.06

M.S. · Computer Science and Technology | Advisor: Prof. Liu Qian

· Dexterous-hand tactile perception and manipulation, reinforcement learning and imitation learning.

Hebei University of Technology (211) 2020.09 – 2024.06

B.Eng. · Internet of Things Engineering

PROFESSIONAL EXPERIENCE

AgiBot 2026.07 – Present

Robot Motion Intelligence Intern Shanghai

- **Project 1 · Real-robot RL** Built a **HIL-SERL** real-robot RL pipeline for dexterous manipulation (O10hand + Franka Research 3); profiled the initial demo data to set the initial exploration-domain bounds, then grew the bounds sinusoidally during training, reaching **99%** success within **20 min**
- **Project 2 · real2sim2real & sim2sim** Built the RL training pipeline for the company flagship high-DoF hand; analyzed reward dynamics for **long-horizon credit assignment**; delivered simulation setup, training, sim-to-sim transfer, **system identification** and **policy deployment**
- **Project 3 · Illumination-robust HIL-SERL** With a **world model**, decoupled environment appearance from proprioceptive state and re-rendered the data, raising lighting robustness from **65.7%** to over **90%** and recovering to **99%** with light human intervention
- **Project 4 · SuperMimic reproduction with a PPO baseline** Uses a differentiable world model instead of large-scale simulation sampling: BPTT over short imagined rollouts inside the model yields analytic gradients, not PPO-style environment interaction; moved the flagship hand's real2sim2real demo onto it, trained the policy without demonstration data, and benchmarked it against PPO on sample efficiency, convergence speed and final success rate

RESEARCH EXPERIENCE

Research on Visuo-Tactile Fusion for In-Hand Dexterous Manipulation 2025.09 – 2026.01

Master's student · Advisor: Prof. Liu Qian Wireless Multimedia Technology Lab, Dalian University of Technology

- **Paper**: Sole first author in **Frontiers in Robotics and AI** (JCR Q2), 2026.01 — **TouchWGNN: spatio-temporal tactile perception for multimodal dexterous manipulation**
- **Sensor layer**: Designed a piezoresistive tactile sensing array for a five-fingered dexterous hand and its calibration method, turning raw array signals into a tactile representation usable by policy learning
- **Perception layer**: A hierarchical GNN models the finger–contact spatial topology of in-hand touch; a dual-branch high-/low-frequency Transformer captures transient micro-vibrations and steady-state contact forces separately
- **Multimodal fusion**: Cross-attention with frequency-domain gating fuses the asynchronous timing of online visual perception and high-speed tactile servoing — vision supplies the goal, touch supplies the contact state
- **RL and Sim2Real**: Large-scale parallel simulation in Isaac Gym with PPO for in-hand manipulation; grid-searched actuator joint responses for system identification, plus domain randomization to narrow the Sim2Real gap
- **Real-robot deployment**: A real-time tactile acquisition, filtering and force-calibration chain on ROS 2 Humble with a Linux RT kernel for high-frequency control; the policy ran closed-loop on a real five-fingered hand

SELECTED PROJECTS

VLA Post-Training with Offline Policy Optimization for Bimanual Garment Folding 2026.09 – Present

- **Task and base model** Folded 4 garment categories (long/short sleeves, trousers/shorts) with tianji bimanual dexterous hands; post-trained the Pi0.5 VLA base with multi-view visual and proprioceptive inputs and action-chunk outputs
- **Offline policy optimization (AWR + RECAP)** The flow-matching head gives only a velocity field with no action density, so log-prob on-policy methods do not apply: AWR reweights data by exp(advantage) under a KL constraint, and RECAP adds dual-mode training — an unconditional pass learns the original distribution, while an advantage-token conditional pass regresses only high-advantage actions and is the sole deployment mode
- **Reward and advantage estimation** Built dense intermediate rewards from keypoint checkpoints, reused the success-probability head as a value function and ran multi-step advantage estimation with GAE, turning sparse success into optimizable per-segment advantages
- **Lightweight world model** Added multi-task auxiliary heads predicting keypoint distances several steps ahead; keypoint-distance sequences replace pixel reconstruction as the dynamics model, giving look-ahead signals at no extra inference cost
- **Distributed asynchronous training pipeline** Decoupled training and rollout nodes via a model/data repository for asynchronous, independently scalable execution; combined domain randomization (texture/lighting/camera/physics), curriculum learning, success replay and hard-example mining
- **Sim2Real and human-in-the-loop** Sim-to-real via behavior-cloning fine-tuning and multi-source data mixing, with camera-pose and speed alignment plus image augmentation; DAgger collected short corrective segments at failures, adding error recovery

SKILLS & HONORS

Technical Skills: `Robot platforms: Newton, ROS 2, MuJoCo, Isaac Gym, LeRobot, MjLab`

`Learning & deployment: PyTorch, Jax, PPO/SAC/RLPD, ACT, Diffusion Policy, VLA, LoRA, Flow Matching, AWR/RECAP, world models, Sim2Real, Sim2Sim`

`Tactile: tactile perception, visuo-tactile fusion, object-centric, GNN`

`Systems: ZeroMQ, WebSocket, Linux RT, shared-memory IPC`

Languages: `CET-6`

Awards: `Provincial First Prize, Math Competition` `National Third Prize, English Competition`

`Academic Excellence Scholarship` `Merit Student`